

**** Autonomous Indoor Drone Navigation ****

Installation Guide

This guide explains how to set up the complete simulation environment for the Autonomous Indoor Drone Navigation project using PX4 Autopilot, ROS 2 Humble, Gazebo Sim (Garden), and Micro XRCE-DDS Agent on Ubuntu 22.04 LTS.

Table of Contents

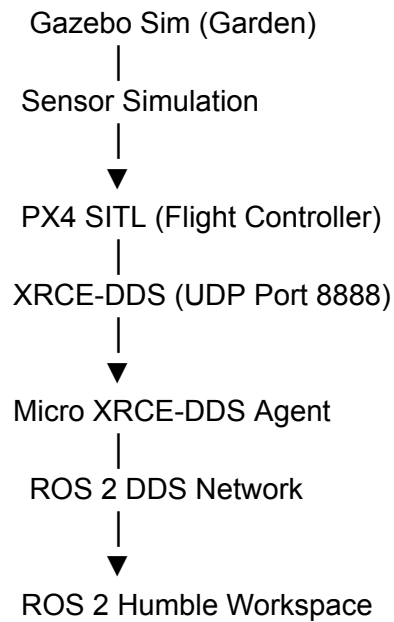
- System Requirements
 - Software Stack
 - 1. Install Ubuntu 22.04 LTS
 - 2. Install ROS 2 Humble
 - 3. Install Colcon & ROS Tools
 - 4. Install PX4 Autopilot v1.14
 - 5. Install Gazebo Sim (Garden)
 - 6. Install Micro XRCE-DDS Agent
 - 7. Install PX4 ROS 2 Messages
 - Useful Commands
 - Troubleshooting
-

System Requirements

Component	Version
Operating System	Ubuntu 22.04.5 LTS (Jammy Jellyfish)
ROS Distribution	ROS 2 Humble Hawksbill
Flight Controller	PX4 Autopilot v1.14
Simulator	Gazebo Sim (Garden)

Communication Bridge	Micro XRCE-DDS Agent v2.4.2
Build System	Colcon + CMake

Software Stack



-
1. Install Ubuntu 22.04 LTS version

Check Version

```
lsb_release -a
```

```
Distributor ID: Ubuntu
Description:   Ubuntu 22.04.5 LTS
Release:      22.04
Codename:     jammy
```

2. Update System

```
sudo apt update
sudo apt upgrade -y
sudo apt install git curl wget build-essential cmake python3-pip python3-colcon-common-extensions -y
```

3. Install the ROS2 distribution [source](#)

Check Version

```
echo $ROS_DISTRO
Humble
```

4. Install Colcon + ROS Tools

```
sudo apt install python3-colcon-common-extensions python3-rosdep python3-vcstool -y
sudo rosdep init
rosdep update
```

4. Install PX4 Autopilot v1.14 [source](#)

```
cd ~
git clone -b v1.14.0 --recursive https://github.com/PX4/PX4-Autopilot.git
cd ~/PX4-Autopilot
bash ./Tools/setup/ubuntu.sh --no-sim-tools
```

Remember to **reboot** your machine after this script runs for the first time.

Build PX4 (After reboot):

```
cd ~/PX4-Autopilot
make px4_sitl
```

Check Version

```
cd ~/PX4-Autopilot
git describe --tags
```

v1.14.0

5. Install gazebo [source](#)

```
sudo apt update && sudo apt install -y curl lsb-release gnupg
sudo wget https://packages.osrfoundation.org/gazebo.gpg -O /usr/share/keyrings/pkgs-osrf-archive-keyring.gpg
echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/pkgs-osrf-archive-keyring.gpg]
http://packages.osrfoundation.org/gazebo/ubuntu-stable $(lsb_release -cs) main" | sudo tee
/etc/apt/sources.list.d/gazebo-stable.list > /dev/null
```

```
sudo apt update
sudo apt install gz-garden
sudo apt install ros-humble-ros-gz-garden
```

```
echo "export GZ_VERSION=garden" >> ~/.bashrc
source ~/.bashrc
```

```
sudo apt update && sudo apt install -y \
ros-humble-desktop \
ros-humble-ros-gz-garden-bridge \
ros-humble-octomap \
ros-humble-octomap-server \
ros-humble-octomap-msgs \
ros-humble-octomap-ros \
ros-humble-octomap-rviz-plugins \
ros-humble-pcl-ros
```

Check Version

```
gz sim --version
Gazebo Sim, version 7.9.0
Copyright (C) 2018 Open Source Robotics Foundation.
Released under the Apache 2.0 License.
```

6. Micro XRCE-DDS Agent Installation [source](#)

```
git clone -b v2.4.2 https://github.com/eProsima/Micro-XRCE-DDS-Agent.git
cd Micro-XRCE-DDS-Agent
mkdir build
cd build
cmake ..
make
sudo make install
sudo ldconfig /usr/local/lib/
```

Check Version

```
cd ~/Micro-XRCE-DDS-Agent  
git describe --tags
```

v2.4.2

7. Install PX4 ROS2 Messages

```
mkdir -p ~/px4_ros2_ws/src  
cd ~/px4_ros2_ws/src  
git clone https://github.com/PX4/px4_msgs.git -b release/1.14  
cd ~/px4_ros2_ws  
source /opt/ros/humble/setup.bash  
colcon build
```

- After building, source before running any code

```
source ~/px4_ros2_ws/install/setup.bash
```

- Otherwise

```
echo "source ~/px4_ros2_ws/install/setup.bash" >> ~/.bashrc  
source ~/.bashrc
```

8. If you want to rebuild colcon after some changes in file

```
cd ~/px4_ros2_ws  
colcon build --symlink-install  
source install/setup.bash
```

9. Kill all the background command

```
killall -9 px4  
killall -9 ruby  
killall -9 MicroXRCEAgent  
pkill -9 -f "gz sim"  
pkill -9 -f "parameter_bridge"
```